EXPERIMENT 8

Interactive dashboards with tableau

Gayathri M
241801071

1. AIM

To analyse the Sample Superstore dataset using Python, Pandas, Matplotlib and Seaborn, and visualize sales and profit across product categories, regions and months.

2. LEARNING OBJECTIVES

• Load a CSV dataset and inspect its rows, columns, data types and missing values.

• Summarize the dataset using descriptive statistics.

• Group and aggregate sales and profit by category, region and month.

• Create bar charts, line charts and scatter plots to understand business trends.

• Use filtering functions to analyse a selected region or product category.

3. BUILT-IN METHODS / FUNCTIONS USED

Method / Function	Purpose
pd.read_csv()	Reads the CSV dataset into a DataFrame.
df.head(), df.shape, df.columns	Displays sample rows, dataset dimensions and column names.
df.info(), df.isnull().sum()	Shows data types and counts missing values.
df.describe()	Generates descriptive statistical summaries.
pd.to_datetime()	Converts order dates into date-time values.
groupby(), sum(), sort_values(), reset_i	ndGerxo(u) ps records and calculates aggregated sales/profit.
plt.figure(), plt.title(), plt.xlabel(), plt.ylab	Displays matplotlib charts
sns.barplot(), sns.lineplot(), sns.scatter	loCt(r)eates bar, line and scatter visualizations using Seaborn.
round(), iterrows(), .dt.to_period()	Rounds values, iterates through rows and extracts monthly periods.

4. PROGRAM (CONTINUOUS FORMAT)

```
import pandas as pd import
matplotlib.pyplot as plt import
seaborn as sns


 df = pd.read_csv("/content/Sample - Superstore.csv",
encoding="latin1") print("Dataset loaded successfully!") print("Dataset
Shape:", df.shape) print("\nFirst 5 rows:") print(df.head())
print("\nColumn Names:") print(df.columns) print("\nDataset
Information:") df.info() print("\nMissing Values:")
print(df.isnull().sum()) print("\nStatistical Summary:")
print(df.describe())
```

```
   df["Order   Date"]   =   pd.to_datetime(df["Order   Date"])   category_sales   =
  df.groupby("Category")["Sales"].sum().sort_values(ascending=False)   print("\nSales
  by Category:") print(category_sales)
plt.figure(figsize=(8, 5))
  sns.barplot(x=category_sales.index,
  y=category_sales.values) plt.title("Sales by Category")
  plt.xlabel("Category") plt.ylabel("Total Sales")
  plt.tight_layout() plt.show()


category_region = df.groupby(["Category", "Region"])["Sales"].sum().reset_index()
  plt.figure(figsize=(10, 6)) sns.barplot(data=category_region, x="Category",
  y="Sales", hue="Region") plt.title("Sales by Category and Region")
  plt.xlabel("Category") plt.ylabel("Sales") plt.tight_layout() plt.show()


df["Month"] = df["Order Date"].dt.to_period("M").astype(str) monthly_sales = df.groupby(["Month",
  "Region"])["Sales"].sum().reset_index()                plt.figure(figsize=(12,                6))
  sns.lineplot(data=monthly_sales, x="Month", y="Sales", hue="Region", marker="o") plt.title("Sales
  Trend    Over    Time")    plt.xlabel("Month")    plt.ylabel("Sales")    plt.xticks(rotation=45)
  plt.tight_layout()   plt.show()   scatter_data   =   df.groupby(["Sub-Category",   "Category"],
  as_index=False)[["Sales",  "Profit"]].sum()  print("\nProfit  vs  Sales:")  print(scatter_data)
  plt.figure(figsize=(10, 7))
  sns.scatterplot(data=scatter_data, x="Sales", y="Profit", hue="Category", s=150)
  for _, row in scatter_data.iterrows():
      plt.text(row["Sales"], row["Profit"], row["Sub-Category"], fontsize=8)
  plt.title("Profit vs Sales by Sub-Category")
  plt.xlabel("Sales") plt.ylabel("Profit")
  plt.axhline(0, linestyle="--")
  plt.tight_layout() plt.show()



  summary = df.groupby("Category")[["Sales", "Profit"]].sum()
  print("\nSales and Profit by Category:") print(summary)
def filter_by_region(region):
      filtered_df = df[df["Region"] == region] print("\nSelected Region:", region)
      print("\nSales                    by                    Category:")
      print(filtered_df.groupby("Category")["Sales"].sum().sort_values(ascending=Fals
      e))  print("\nTotal Sales:", round(filtered_df["Sales"].sum(), 2)) print("Total
      Profit:", round(filtered_df["Profit"].sum(), 2)) filter_by_region("East")

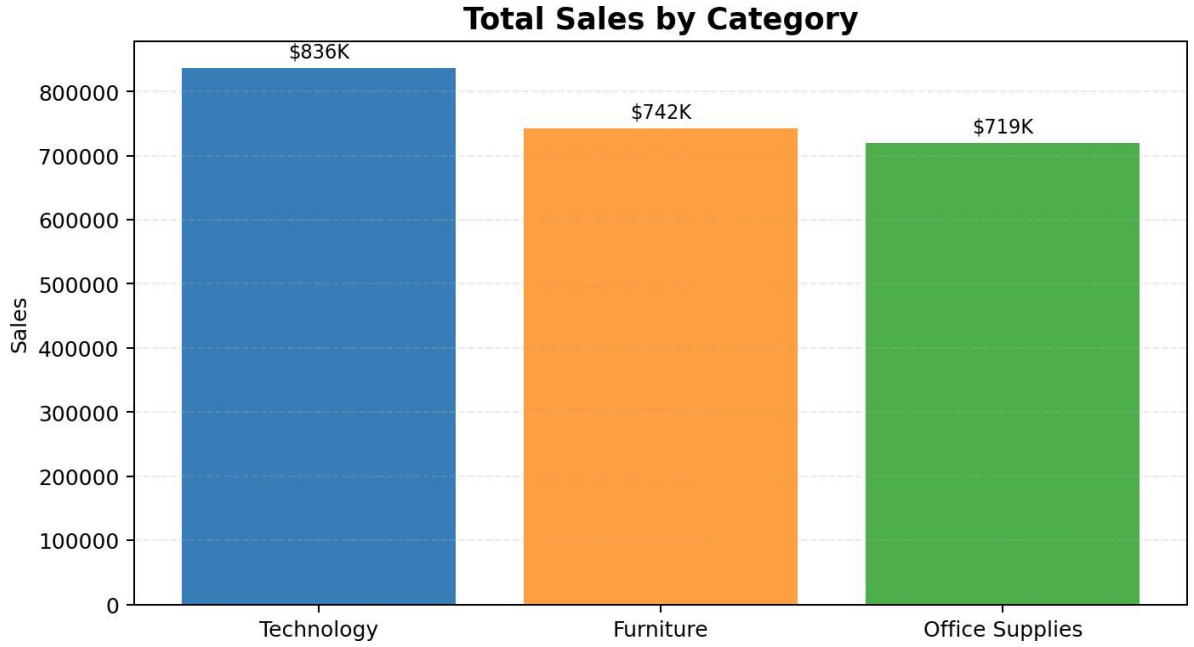

  # Filter and analyse a selected category def
  filter_by_category(category):
      filtered_df = df[df["Category"] == category] print("\nSelected Category:",
  category) print("Total Sales:", round(filtered_df["Sales"].sum(), 2)) print("Total
  Profit:", round(filtered_df["Profit"].sum(), 2)) print("\nSales by Region:")
  print(filtered_df.groupby("Region")["Sales"].sum().sort_values(ascending=False))
  filter_by_category("Technology")

  # Overall dashboard summary total_sales = df["Sales"].sum() total_profit
  = df["Profit"].sum()  print("\n" +  "="  * 60)  print("SALES  AND  PROFIT
  DASHBOARD   SUMMARY")   print("="   *   60)   print("Total   Sales   :",
  round(total_sales, 2))  print("Total  Profit:",  round(total_profit,  2))
  print("\nTotal              Sales              by              Region:")
  print(df.groupby("Region")["Sales"].sum().sort_values(ascending=False))
  print("\nTotal             Sales             by             Category:")
  print(df.groupby("Category")["Sales"].sum().sort_values(ascending=False)
  )          print("\nTotal          Profit          by          Category:")
  print(df.groupby("Category")["Profit"].sum().sort_values(ascending=False
  ))
```

5. OUTPUT (COLOUR DIAGRAMS AND RESULTS)

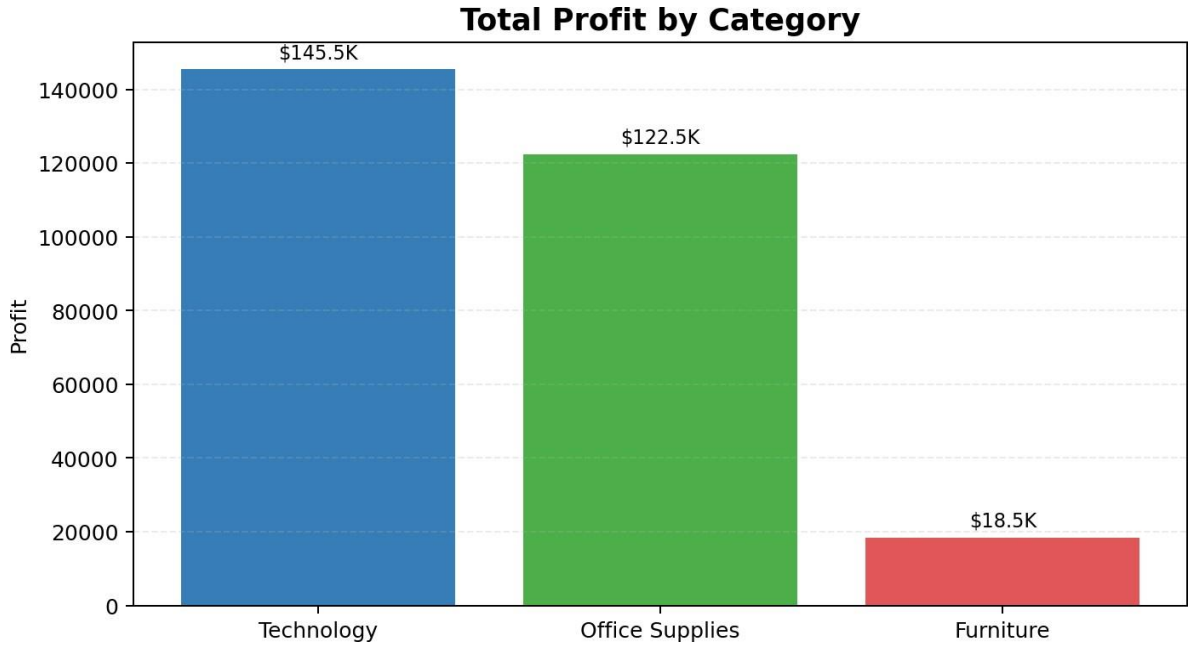
Dataset loaded successfully. Dataset shape: **(9994, 21)**. The dataset contains 9,994 records and 21 columns, and the uploaded output reports no missing values.

Sales by Category



Technology has the highest total sales, followed by Furniture and Office Supplies.

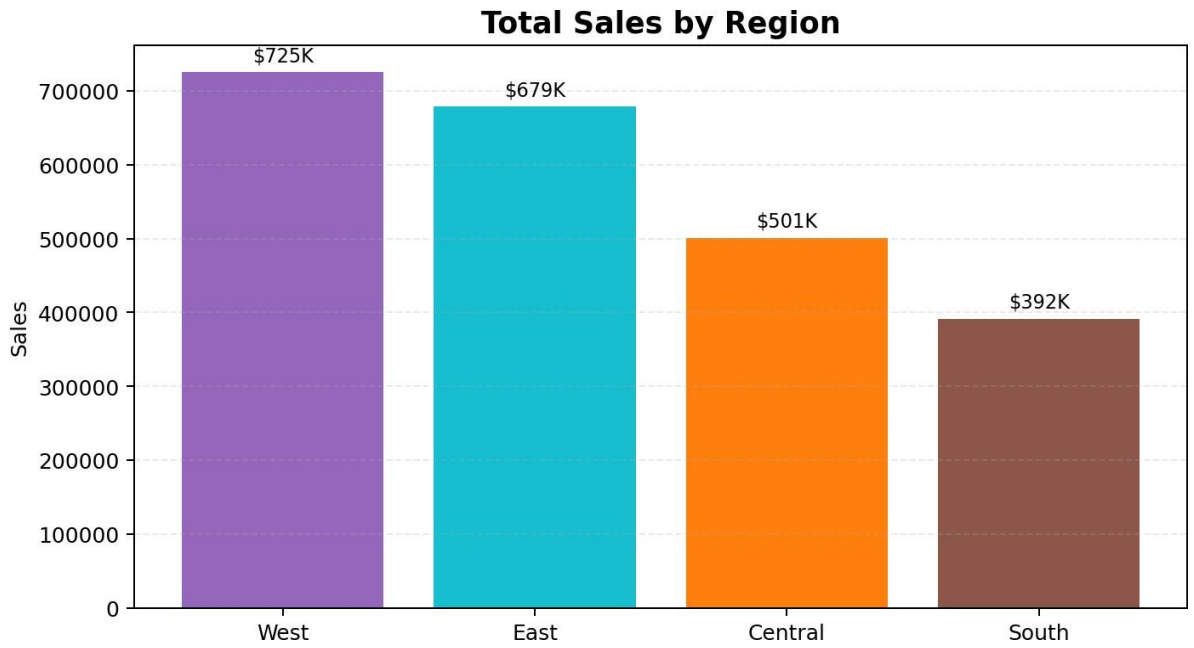
Profit by Category



Technology produces the highest total profit. Furniture has a much smaller profit compared with the other categories.

5. OUTPUT (CONTINUED)

Sales by Region



Overall dashboard summary from the provided output:

Measure	Value
Total Sales	$2,297,200.86
Total Profit	$286,397.02
Highest Sales Region	West — $725,457.82
Highest Sales Category	Technology — $836,154.03
Technology Profit	$145,454.95
East Region Sales	$678,781.24
East Region Profit	$91,522.78

6. LEARNING OUTCOMES

• Loaded and explored the Superstore dataset using Pandas.

• Checked data types, missing values and statistical summaries.

• Used grouping and aggregation to compare sales and profit.

• Visualized category, region and monthly trends with colourful charts.

• Created reusable filtering functions for region-wise and category-wise analysis.

RESULT

Thus, the Superstore dataset was analysed successfully using Python. The charts and summary statistics show that Technology generated the highest sales and profit, while the West region recorded the highest sales.